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19th Jan 2021 | DAG 360

→ Optimization or Mathematical Program

- Age of Tom is 30 & age of Tom is 5 times age of Jerry. Find Jerry's age.

$J \qquad T = 30$

$T = 5 \times J \Rightarrow J = \frac{30}{5} = 6$ } 1 variable, 1 eq^n

- Age of Tom is 4x age of Jerry. The diff. btw. the age of tom & jerry is 30. Find age of Tom & Jerry.

$T = 4 \times J \qquad 4J - J = 30$

$T - J = 30 \qquad 3J = 30 \qquad J = 10$

$T, J \geq 0 \qquad T = 40$ } 2 variables, 2 eq^ns

- We have ₹100/-, Potato ₹20/kg, Tomato ₹25/kg

Buy as much as possible (in terms of wt in kg)

x : amt. of potato

y : amt. of tomato

max $x + y \Rightarrow$ objective

s.t. $\underbrace{20x + 25y \leq 100, \quad x \geq 0, \quad y \geq 0}_{\text{constraints}}$

Q: Express this diagrammatically.

- Let's say we have a bag with max. capacity of 5 kg. Fill the bag fully by spending as low as possible.

$$\min\ 20x + 25y$$

$$x + y = 5, \quad x \geq 0, \quad y \geq 0$$

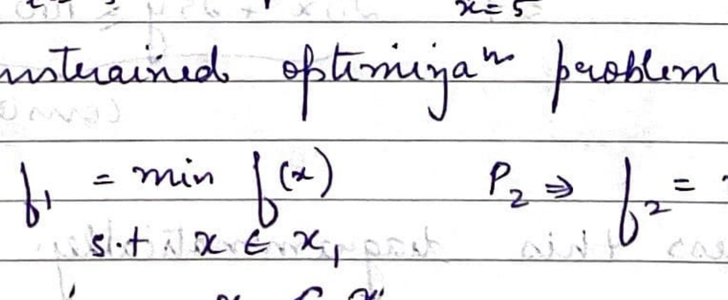
→ Standard Formulaⁿ

min or max / arg min or arg max $f(x)$ → objective funcⁿ

s.t. $x \in \mathcal{X}$ → constraint set

$$x = \begin{bmatrix} x(1) \\ \vdots \\ x(d) \end{bmatrix} \in \mathbb{R}^d$$

- $\min\ f(x) = x^2$ s.t. $x \geq 5$



$f_x = 25$
$x = 5$

Constrained optimizaⁿ problem.

$P_1 \Rightarrow f_1 = \min f(x)$ s.t. $x \in \mathcal{X}_1$ $\qquad$ $P_2 \Rightarrow f_2 = \min f(x)$ s.t. $x \in \mathcal{X}_2$

$$\mathcal{X}_1 \subseteq \mathcal{X}_2$$

$$f_1 \geq f_2$$

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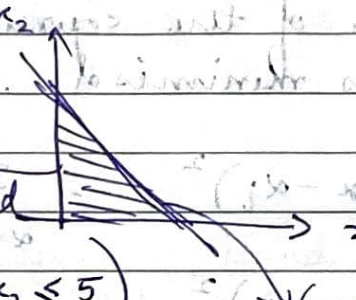
std. formula^n

$$\text{s.t.}\quad g_1(x(1), \ldots x(d)) \le c_1$$
$$\vdots$$
$$g_m(x(1), \ldots x(d)) \le c_m$$
} inequality constraints

$$l_1(x(1), \ldots x(d)) = 0$$
} equality constraints

min. $20x_1 + 25x_2$ — $x_1 \ge 0$, $x_2 \ge 0$

$x_1 + x_2 = 5$



- Goal of the optimiza^n –

1. formulate the optimiza^n problem (P)
2. Categorize P → properties of f & x
3. identify a suitable algorithm to solve it.
4. analyze the algo.
5. Try to find $P' \equiv P$, $P' \simeq P$ because P' is easier to solve.
6. perform the sensitivity analysis, if we perturb the 'P', then what happens

interchangeable

- Properties of f → continuity, convexity, differentiability.

how well-behaved or easy the objective func^n is.

- Properties of $x \Rightarrow$ convexity, affine